

Dhyanam Mahajan

Software Engineer | Resume

Key Experience & Projects

- Built an object detection pipeline using YOLOv8, PyTorch, and rasterio to process GeoTIFF drone images with sliding-window tiling.
- Fine-tuned Llama 3 8B model using LoRA and PyTorch for domain-specific text classification, NLP, and automated summarization.
- Designed and scaled backend microservices using FastAPI, PostgreSQL, and Redis caching for real-time notifications and high-throughput REST APIs.
- Built a Real-Time Voice AI Assistant using Silero VAD, whisper.cpp, WebRTC, audio processing, and Python.
- Developed a full-stack dashboard using React, TypeScript, Node.js, Express, GraphQL, and Tailwind CSS with interactive state management.